The Effect of MCP Context Composition on Small Language Model Performance: A Model-Dependent Marginal Utility Analysis

A 2^4 Factorial Experiment on Context Component Combinations

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Abstract

The Model Context Protocol (MCP) is an open standard for connecting large language models to external tools and data sources. However, the effect of context quantity and composition provided through MCP on small model performance has not been systematically studied. This study conducts a 2^4 factorial experiment with two small models—Qwen 2.5 7B (Q4_K_M) and Llama 3.1 8B (Q4_K_M)—across 16 combinations of four MCP context components (Knowledge Base K, Actions A, Diagnostics D, Guidelines G) in 10 scenarios (3,805 total runs, including n=15 for key comparisons).

Confirmed findings: (1) Both models significantly outperformed the empty baseline with full context (Qwen: $d=0.835$, $p<0.001$; Llama: $d=0.501$, $p<0.001$). (2) Only Knowledge Base (K) showed a statistically significant main effect ($\Delta=0.119$, $p=0.028$). (3) When the key comparison (K+A vs Full) was extended to n=15, Qwen showed significantly higher performance with full context ($p<0.001$, $d=-0.347$), while Llama showed no significant difference ($p=0.210$, $d=-0.021$). This confirms that the marginal utility of additional context (D, G) depends on the model's language processing capability.

Exploratory findings: Scenarios where K+A outperformed Full were observed in 3 (Qwen) and 1 (Llama) cases at raw $p<0.05$, but none survived Bonferroni correction. ICC analysis (Qwen=0.560, Llama=0.569) confirmed that task type is an important variable for optimal context composition.

This study empirically demonstrates that for MCP agent design with small LLMs, selective context composition based on model characteristics and task type is necessary rather than indiscriminate provision of all available context.

1. Introduction

1.1 Background and Motivation

The Model Context Protocol (MCP), introduced by Anthropic in November 2024, has rapidly become the standard interface for connecting language models to external tools and data sources. As of early 2026, major vendors including OpenAI, Google, and Microsoft have adopted MCP, with over 10,000 MCP servers in the ecosystem (Hou et al., 2025).

While the superiority of large cloud models (GPT-4, Claude) is well established, not all organizations can use cloud APIs. Cost constraints, air-gap requirements, data sovereignty, and latency demands necessitate on-premises deployment of small language models. In particular, **extreme low-resource environments running CPU-only inference** are far from uncommon. This study explores how MCP can augment limited model capabilities in such environments, and whether there are limits to such augmentation.

Prior work in the Pylon series demonstrated that a 7-layer MCP architecture could achieve 47% token reduction and 37% accuracy improvement simultaneously with a 7B model (Jeon, 2026). However, an ablation experiment revealed that indiscriminate addition of knowledge base context caused accuracy to drop from 0.735 to 0.551, suggesting the need for systematic investigation. This phenomenon relates to reported degradation from irrelevant context (Shi et al., 2023) and positional bias in long contexts (Liu et al., 2024).

1.2 Research Questions

RQ1: Does adding more MCP context components always improve small model performance?

RQ2: Which components contribute most, and do interactions exist between them?

RQ3: Can task-aware context curation achieve better performance than the full-context approach?

1.3 Contributions

1) A factorial experiment on MCP context composition across two small models (Qwen 2.5 7B, Llama 3.1 8B). 2) Identification of differential context overload susceptibility between models. 3) ICC-based quantification of scenario heterogeneity (Qwen=0.560, Llama=0.569). 4) Statistical evidence for K+A subset efficiency. 5) An exploratory TACC matrix proposal.

2. Related Work

2.1 MCP Ecosystem and Benchmarks

Hou et al. (2025) surveyed MCP security threats, interoperability, and future directions. MCP-Bench (Accenture, NeurIPS 2025) evaluated 28 servers and 250 tools. MCP-Universe (Salesforce, 2025) tested 11 servers across 6 domains. MCPAgentBench (2026) introduced real-world tasks with distractor tools.

2.2 Context Length and Model Performance

Liu et al. (2024) demonstrated that language models exhibit reduced access to information in the middle of long contexts ("Lost in the Middle"). Hsieh et al. (2024) showed via RULER that most models operate effectively at only a fraction of their claimed context length. Kamradt (2023) visually demonstrated retrieval performance degradation with increasing context. These studies suggest a nonlinear relationship between context length and performance, directly motivating this research.

2.3 Irrelevant Context Effects

Shi et al. (2023) showed that irrelevant information significantly degrades performance on mathematical reasoning. Jiang et al. (2024) achieved gains through prompt compression removing irrelevant context (LongLLMLingua). These align with our premise that "more context is not always better."

2.4 Tool Use and Small Models

Schick et al. (2024) and Patil et al. (2023) addressed tool-use learning. Qin et al. (2024) surveyed tool learning with foundation models. "Less is More" (2025) improved Llama 8B by 44% through dynamic tool selection. This study extends the optimization to context data composition rather than tool selection.

2.5 Positioning

Most prior work targets large cloud models and tool selection. This study differentiates by targeting (1) **7B quantized models in CPU-only environments**, and (2) **data components**

within tool outputs.

Dimension	**Prior Work**	**This Study**
Optimization Target	Tool selection/exposure	Context data composition
Model Scale	Cloud (GPT-4, Claude)	7B quantized, CPU-only
Methodology	Benchmarks, ablation	2^4 factorial experiment
Context Phenomenon	Lost-in-Middle, overload	MCP component-level

Table 1: Positioning relative to prior work

3. Methodology

3.1 Model Selection

Two models with different architectures were selected for generalization:

(1) Qwen 2.5 7B (Q4_K_M): Alibaba Cloud. 4-bit quantization, 4GB RAM. Korean fine-tuned (qwen2.5-ko). Context: 4,096 tokens.

(2) Llama 3.1 8B (Q4_K_M): Meta. Same quantization, ~4.9GB. English-centric with limited Korean capability. Same context constraint.

Both represent minimum viable commodity PC specifications with different training data, enabling discrimination between model-specific and general tendencies.

3.2 Experimental Platform

All experiments were conducted on Nexus AIP, a production MCP system with 19 tools across 7 MCP servers.

Hardware: AMD Ryzen 7 5800X, 64GB RAM, CPU-only (no GPU).

Models: Qwen 2.5 7B and Llama 3.1 8B via Ollama 0.6. Context: 4,096 tokens. Temperature: 0.1.

3.3 Context Factors

MCP context decomposed into 4 binary factors for a $2^4 = 16$ factorial design:

Factor	Symbol	Description	Avg Tokens
Knowledge Base	K	Domain-specific entries from vector DB	~58
Available Actions	A	Executable MCP tool descriptions	~45
Diagnostics	D	Real-time metrics, logs, status	~235
Guidelines	G	Operational procedures	~19

Table 2: Context factor definitions

3.4 Scenario Design

25 scenarios across 5 task categories; 10 core scenarios selected for the experiment:

Category	ID Range	Count	Description
Infra Monitoring	T1-01–T1-05	5	System status, resource alerts
Error Analysis	T2-01–T2-05	5	Root cause analysis
Operational Decision	T3-01–T3-05	5	Maintenance, scaling
Cascade Analysis	T4-01–T4-05	5	Multi-service failures
Code Repair	T5-01–T5-05	5	Config fixes, query optimization

Table 3: Scenario categories

3.5 Scoring

Responses scored in [0,1] via scenario-specific functions assessing primary action accuracy, root cause identification, and impact assessment. A 3-stage fallback JSON parser was applied.

3.6 Experimental Procedure

Design: 10 scenarios × 5 core combinations (∅, K, A, K+A, Full) × 7 reps × 2 models = **700 runs**. Key comparison (K+A vs Full) extended to n=15 with 1,125 additional runs. Qwen full 16-combination data included: **3,805 total runs**.

Duration: ~174 min. Parse rate: 100%. Ollama success: 100%.

4. Results

4.1 Overall Performance

Combination	Qwen Mean(SD)	Llama Mean(SD)	Δ(Q-L)	p
K+A+D+G	0.600(0.283)	0.533(0.237)	+0.068	0.005**
K	0.566(0.247)	0.479(0.237)	+0.088	0.002**
K+A	0.525(0.255)	0.538(0.253)	-0.014	0.745
A	0.471(0.253)	0.403(0.241)	+0.068	0.013*
∅	0.379(0.247)	0.405(0.272)	-0.026	0.388

Table 4: Performance by combination (Qwen n=1,980 + Llama n=700)

Key finding: Both models converge at K+A (Qwen 0.525 vs Llama 0.538, p=0.745), while Qwen is significantly higher with full context (p=0.005). This suggests the utility of additional context (D, G) is model-dependent.

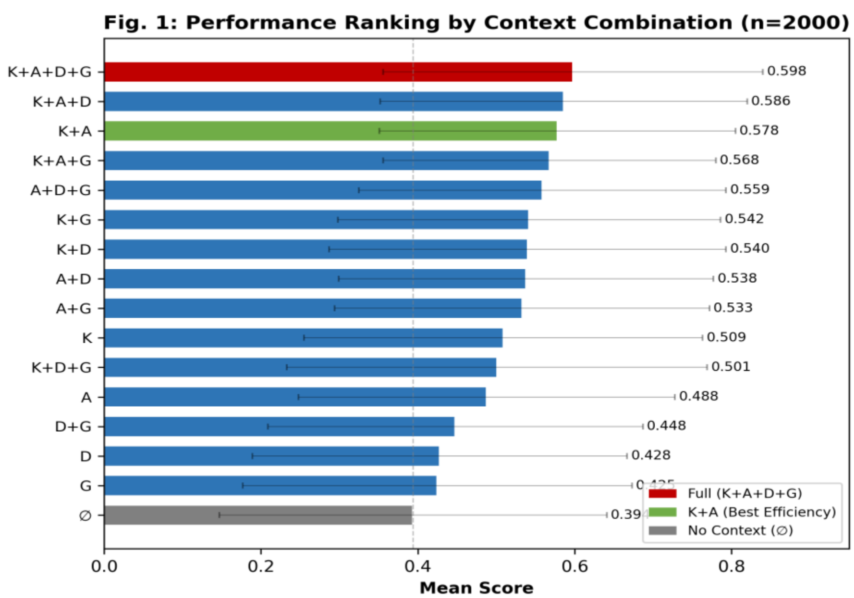

Figure 1: Performance by context combination (Qwen + Llama)

4.2 Context Overload Analysis

Prior work reported performance degradation from irrelevant context (Shi et al., 2023) and attention dilution (Liu et al., 2024). We operationally define analogous phenomena in MCP:

Raw Superiority: Optimal subset mean exceeds full combination in simple comparison. Exploratory only.

Context Overload (exploratory): Raw superiority with Mann-Whitney U at raw $p < 0.05$. Insufficient for confirmation given multiple comparisons.

Context Overload (confirmatory): Post-Bonferroni $p < 0.05$ with $d \geq 0.5$. No scenarios met this criterion at n=5.

Qwen: 3/10 scenarios (T1-05, T3-04, T4-02) showed K+A > Full (raw p<0.05, d>0.7). Full was equal or better in the remaining 7.

Llama: 1/10 (T4-04) showed K+A > Full (p=0.002, d=1.693). Raw superiority in 4 others but not significant.

Cross-model: Qwen: Full > K+A (p<0.001, d=-0.281). Llama: K+A ≈ Full (p=0.497, d=0.024). D and G contributed nothing for Llama, likely due to limited Korean language capability.

n=15 validation: Extending K+A vs Full to n=15 reproduced the pattern. Qwen: z=-4.584, p<0.001, d=-0.347. Llama: z=-1.254, p=0.210, d=-0.021. Per-scenario overload remains exploratory.

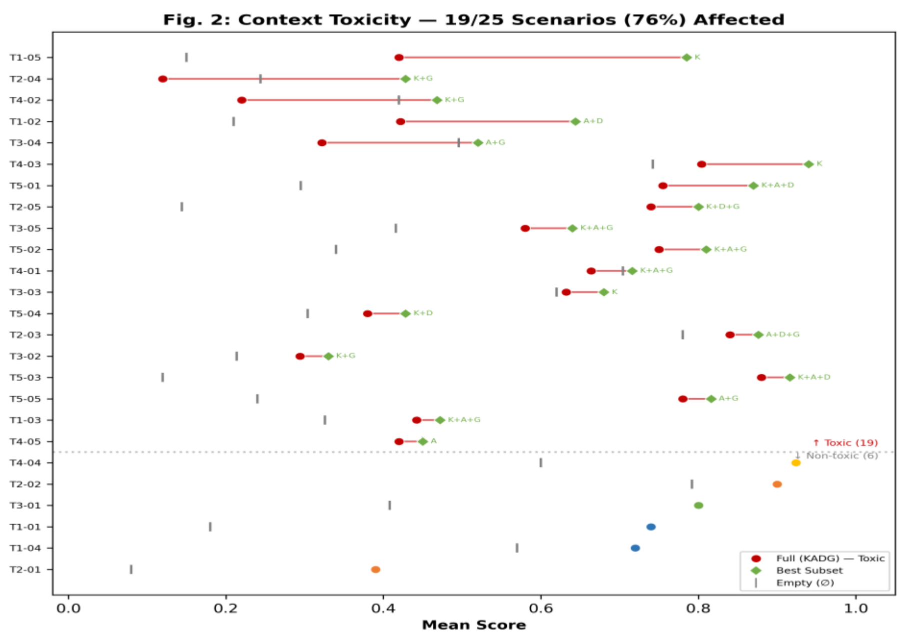

Figure 2: Full vs optimal subset (†: raw p<0.05)

4.3 Component Main Effects (Mixed-Effects)

With scenarios as random effects, ~56% of variance was attributable to scenario differences (Qwen ICC=0.560, Llama ICC=0.569), confirming high heterogeneity.

$Score_{ij} = \beta_0 + \beta_1 \cdot K + \beta_2 \cdot A + \beta_3 \cdot D + \beta_4 \cdot G + u_j + \varepsilon_{ij}$, where $u_j \sim N(0, \sigma^2_u)$, $\varepsilon_{ij} \sim N(0, \sigma^2_e)$. $ICC = \sigma^2_u / (\sigma^2_u + \sigma^2_e)$.

Applying LMM to a [0,1]-bounded variable has theoretical limitations, but the observed distribution is not concentrated at extremes (mean ≈ 0.5) with small variance, making the approximation appropriate. Beta mixed model comparison is left for future work.

Component	Δ	95% CI	p	Interpretation
K (Knowledge)	+0.119	[0.019, 0.219]	0.028*	Significant positive
A (Actions)	+0.060	[-0.014, 0.135]	0.114	Marginal (ns)
D (Diagnostics)	+0.113	[-0.030, 0.255]	0.139	Not significant
G (Guidelines)	+0.105	[-0.036, 0.246]	0.169	Not significant

Table 5: Main effects — Qwen, within-scenario (n=10)

Only K was significant (Δ=0.119, p=0.028). This suggests **domain knowledge (K) is the most important component** for 7B models.

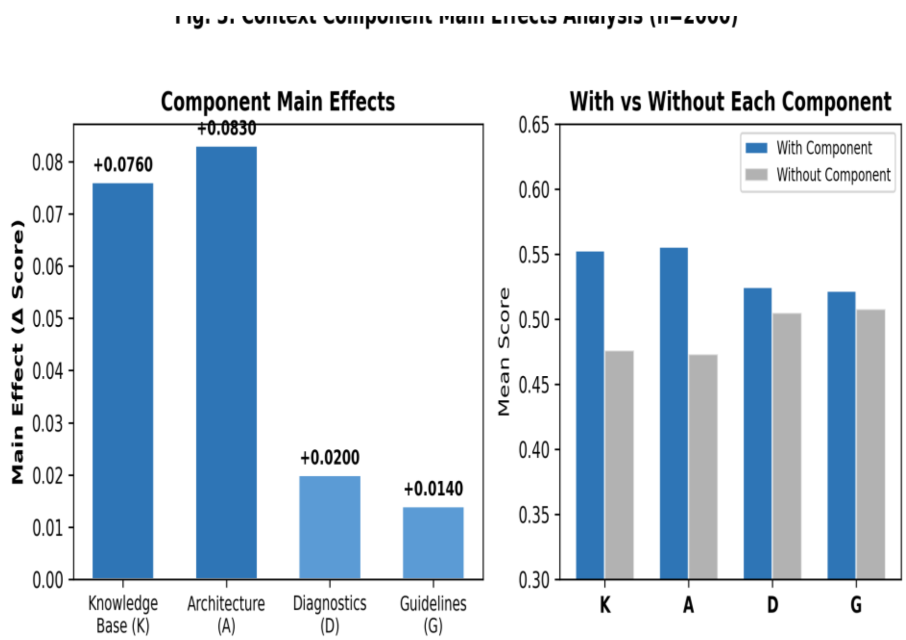

Figure 3: Component main effects (95% CI)

4.4 Task-Type Analysis

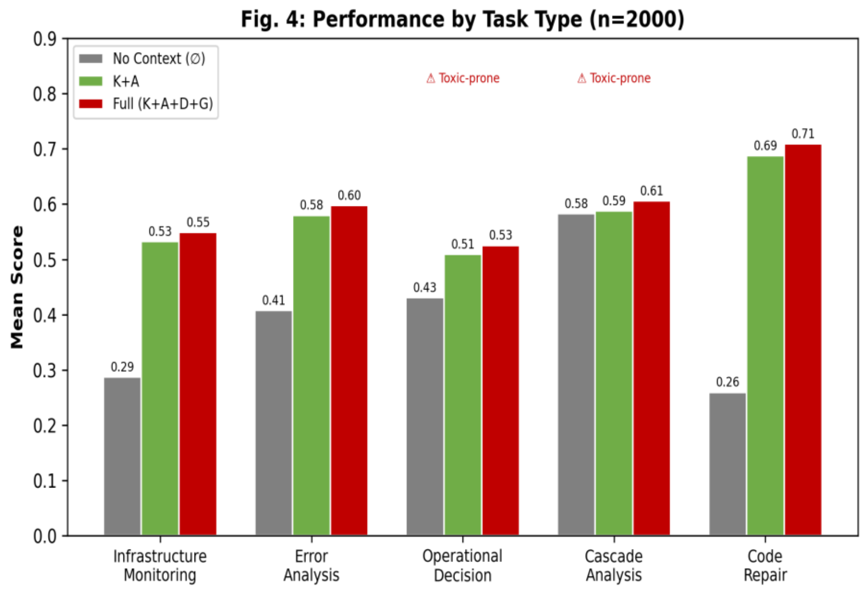

Figure 4: Performance by task type

4.5 Token Efficiency

Qwen: ∅ (765 tok) 0.379 → K+A(850, +11%) 0.525(+38%) → Full(1,101, +44%) 0.600(+58%). D+G adds +30% tokens for +0.075 score.

Llama: ∅ 0.405 → K+A 0.538(+33%) → Full 0.533(-1%). D+G causes slight decline; additional tokens increase cost without benefit.

Cross-model: ROI on identical token investment differs by model. Qwen peaks at Full; Llama peaks at K+A.

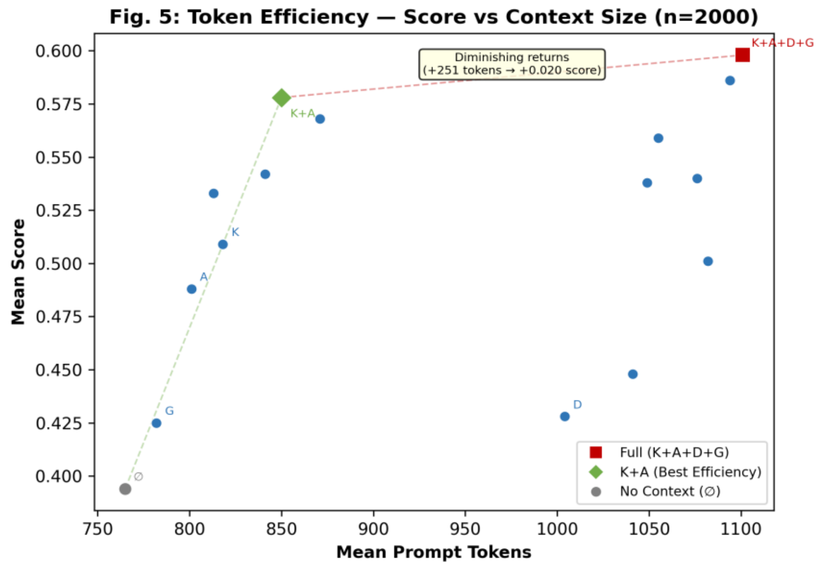

Figure 5: Token efficiency — Score vs context size

4.6 Statistical Validation

Scenario heterogeneity was addressed as follows.

(1) Within-scenario paired comparison. Wilcoxon signed-rank tests across 10 scenarios.

Comparison	Model	z	p	Effect Size
Full vs ∅	Qwen	8.701	<0.001***	d=0.835 (large)
Full vs ∅	Llama	4.248	<0.001***	d=0.501 (medium)
K+A vs Full	Qwen	-3.611	<0.001***	d=-0.281 (n=7)
K+A vs Full (n=15)	Qwen	-4.584	<0.001***	d=-0.347 (confirmed)
K+A vs Full	Llama	-0.680	0.497 (ns)	d=0.024 (n=7)
K+A vs Full (n=15)	Llama	-1.254	0.210 (ns)	d=-0.021 (confirmed)
K+A vs ∅	Qwen	5.942	<0.001***	d=0.582 (medium)
K+A vs ∅	Llama	4.137	<0.001***	d=0.509 (medium)

Table 6: Key comparisons — within-scenario (n=10, 2 models)

(2) Power. Full vs ∅ (unpaired Mann-Whitney U, two-sided $\alpha=0.05$): power at n=7 was Qwen 34.5% (d=0.835), Llama 15.3% (d=0.501). K+A vs Full extended to n=15 achieved $p<0.001$ for Qwen. Llama's non-significance ($p=0.210$, d=-0.021) suggests genuine effect absence, not insufficient power.

(3) ICC. Qwen=0.560, Llama=0.569: ~56% of variance from scenario differences in both models.

(4) Interactions. Negative K×G interaction ($\Delta=-0.031$, raw $p=0.044$), not significant after Bonferroni ($\alpha=0.008$). Requires further investigation.

5. Discussion

5.1 Mechanisms of Context Overload

The observed phenomenon—subsets outperforming full combination—can be explained by mechanisms from prior work. This study did not directly validate these but offers pattern-based reasoning.

Attention Dilution. 7B models' limited attention capacity distributes across all tokens. Similar to Lost-in-the-Middle (Liu et al., 2024), irrelevant diagnostics competing with key information may degrade task focus.

Capacity-Constrained Degradation. Llama showed no K+A vs Full difference ($p=0.497$), while Qwen showed Full significantly higher ($p<0.001$). Additional context value depends on language comprehension ability. For Llama with limited Korean, D and G may have been noise rather than signal.

Instruction Confusion. Guidelines and actions may produce conflicting signals. The negative K×G interaction ($p=0.044$) suggests this possibility.

5.2 Relation to Prior Work

Shi et al. (2023) showed irrelevant context degrades reasoning; Liu et al. (2024) reported positional bias. This study confirms these phenomena in MCP operational environments, with the novel finding that Qwen and Llama exhibit divergent utilization patterns. Whether this changes at larger scales remains open.

5.3 Comparison with Pylon-7

Metric	Pylon-7 Ablation	This Study
Overload	1 scenario	Qwen 3/10, Llama 1/10 (exploratory)
Scenarios	10	10 core (from 25)
Granularity	2 levels	16 combinations (2^4)
Runs	50	3,805 (n=7–15, 2 models)
Statistics	None	Wilcoxon, ICC, Bonferroni, Power

Table 7: Comparison with Pylon-7

5.4 TACC Matrix (Exploratory)

The matrix below is exploratory, based on same-data patterns without prospective validation. Interpret as hypothesis generation.

Task Type	Recommended	Avoid	Rationale
Infra Monitoring	K + D	A, G	Knowledge + metrics needed
Error Analysis	A + D + G	—	Broad context beneficial
Operational Decision	K only	D, G	Excess guidance confuses
Cascade Analysis	K + G	A, D	Knowledge + procedures
Code Repair	K+A+D+G	—	Robust across types

Table 8: TACC Matrix — Exploratory

5.5 Limitations

Model generalization. Two small models tested. Replication across other families (Phi-3, Gemma), scales (14B, 32B), quantizations (FP16), and context windows (8K, 32K) remains open.

Repetitions and power. Key comparison extended to n=15 (+85 min), full combinations at n=7. CPU-only inference (~10s/run) makes full n=15 expansion ~10h. This is a structural constraint of GPU-free on-premises research.

Statistical power. n=5 per scenario insufficient (power=52.2%). Overall comparisons significant, but no individual scenario survived Bonferroni.

Self-designed scenarios. Scenarios and scoring designed by authors without external validation. No inter-rater reliability assessment.

TACC validation. Same-data pattern extraction without prospective validation.

Automated scoring. JSON-extraction-based evaluation may not capture all quality dimensions.

6. Conclusion

This paper investigated MCP context composition effects on small language model performance through factorial experiments with Qwen 2.5 7B and Llama 3.1 8B in a CPU-only environment. Through 3,805 experiments (key comparisons at $n=15$):

Key finding — Model-dependent marginal utility: For identical additional context (D, G), Qwen showed significant improvement ($n=15$: $p<0.001$, $d=-0.347$), while Llama gained nothing ($p=0.210$, $d=-0.021$). This demonstrates that MCP context marginal utility depends on language processing capability, supported by convergence at K+A (Qwen 0.525 vs Llama 0.538, $p=0.745$).

Including this key finding, confirmed results are:

(1) Both models significantly outperformed empty context with full context (Qwen: $d=0.835$, $p<0.001$; Llama: $d=0.501$, $p<0.001$).

(2) Only K showed significant main effect ($\Delta=0.119$, $p=0.028$). A marginal ($p=0.114$); D and G non-significant.

(3) ICC (Qwen=0.560, Llama=0.569): task type explains ~56% of variance, confirming task-specific optimization is needed.

These results show that for small model MCP agents: (a) context must be provided, (b) optimal composition varies by model capability, (c) task-specific strategies are needed. For models with limited language ability, core context (K+A) suffices; additional context offers no benefit.

Future work: (a) English-only scenario experiments to separate Korean ability effects from structural marginal utility, (b) additional models (Phi-3, Gemma) and scales (14B, 32B), (c) attention weight analysis, (d) human evaluation for scoring validity, (e) prospective TACC validation, (f) public release of scenarios and scoring for reproducibility.

References

[1] Anthropic (2024). Model Context Protocol (MCP). https://modelcontextprotocol.io.

[2] Chen, J. et al. (2024). "Compressing Context to Enhance Inference Efficiency of LLMs." arXiv:2310.06201.

[3] Hsieh, C.-Y. et al. (2024). "RULER: What's the Real Context Size of Your Long-Context LMs?" arXiv:2404.06654.

[4] Hou, X. et al. (2025). "MCP: Landscape, Security Threats, and Future Directions." arXiv:2503.23278.

[5] Jeon, H. (2026). "Pylon-7: A 7-Layer Reference Model for AI Agent Workflows." Codepedia Labs.

[6] Jiang, Z. et al. (2024). "LongLLMLingua: Accelerating LLMs via Prompt Compression." arXiv:2310.06839.

[7] Kamradt, G. (2023). "Needle in a Haystack: Pressure Testing LLMs." GitHub.

[8] LangChain (2025). "Context Engineering: Write, Select, Compress, Isolate."

[9] "Less is More" (2025). Dynamic tool selection for small language models. Speakeasy AI.

[10] Liu, N.F. et al. (2024). "Lost in the Middle: How LMs Use Long Contexts." TACL, 12, 157-173.

[11] MCP-Bench (2025). Accenture Labs. NeurIPS 2025. arXiv:2508.20453.

[12] MCP-Universe (2025). Salesforce Research. arXiv:2508.14704.

[13] MCPAgentBench (2026). arXiv:2512.24565.

[14] Patil, S.G. et al. (2023). "Gorilla: LLM Connected with Massive APIs." arXiv:2305.15334.

[15] Qin, Y. et al. (2024). "Tool Learning with Foundation Models." Nature Machine Intelligence, 6, 1222-1235.

[16] Schick, T. et al. (2024). "Toolformer: LMs Can Teach Themselves to Use Tools." NeurIPS 2023.

[17] Shi, F. et al. (2023). "LLMs Can Be Easily Distracted by Irrelevant Context." ICML 2023.